

ADVERSE DRUG REACTION

IDENTIFICATION BASICS

AI-assisted patient case analysis project

Project ID	TASK1
Topic	Identification and assessment of a suspected adverse drug reaction
Case type	Patient-based pharmacovigilance case scenario

1. Objective

To analyze a patient case systematically and determine whether the reported clinical events are potentially related to a medicine. The assessment considers the patient's symptoms, medical history, treatment timeline, dechallenge information, alternative explanations, and the known safety profile of the suspected drug.

2. Patient Scenario

Patient	Female, 45 years
Presenting complaint	Rash, itching, nausea and dizziness
Suspected medicine	Amoxicillin
Indication	Presumed bacterial respiratory infection
Relevant history	No previously documented drug allergy in the case record; no major chronic illness reported

3. Clinical Timeline

Day	Clinical event	Interpretation
Day 0	Amoxicillin started for suspected bacterial infection.	Baseline before suspected reaction.
Day 2	Patient developed itching and a skin rash; nausea and dizziness were also reported.	Temporal association with drug exposure.
Day 2–3	Symptoms persisted after continued exposure.	Supports the need for clinical assessment and consideration of drug withdrawal.
After withdrawal	For this educational case, symptoms are assumed to improve after the suspected medicine is stopped.	Positive dechallenge supports, but does not prove, causality.

Initial impression: The timing of rash and pruritus after initiation of amoxicillin makes a drug-related reaction clinically plausible. The case requires further assessment before a definitive causal relationship is assigned.

4. ADR Assessment

A structured review is used to distinguish a possible adverse drug reaction from symptoms caused by the underlying illness or another exposure.

A. Symptom Review

Finding	Case observation	ADR relevance
Rash	New skin eruption after medicine initiation.	Compatible with a hypersensitivity-type reaction.
Itching	Pruritus appeared with the rash.	Supports an allergic/hypersensitivity pattern.
Nausea	Reported after treatment began.	Can occur with medicines and has several alternative causes.
Dizziness	Reported during the same period.	Non-specific; requires assessment of hydration, infection, blood pressure and other drugs.

B. Causality Considerations

- 1. Temporal relationship:** The symptoms began after amoxicillin was introduced, which supports a possible relationship.
- 2. Biological plausibility:** Amoxicillin is a beta-lactam antibacterial. Cutaneous hypersensitivity reactions are recognized adverse effects of beta-lactam antibiotics.
- 3. Dechallenge:** Improvement after stopping the suspected drug increases suspicion, although spontaneous recovery and treatment of the underlying illness must also be considered.
- 4. Rechallenge:** Re-exposure should not be performed merely to prove causality, particularly when an allergic reaction is suspected.
- 5. Alternative causes:** Viral infection, food/environmental allergens, dehydration, fever, concomitant medicines and the underlying infection may produce some of the reported symptoms.

C. Provisional Pharmacovigilance Classification

Element	Assessment
Suspected drug	Amoxicillin
Suspected reaction	Cutaneous hypersensitivity-type reaction with pruritus/rash; nausea and dizziness are associated reported symptoms.
Seriousness	Not classified as serious in this educational scenario because no hospitalization, death, life-threatening event, disability or congenital anomaly is described.
Expectedness	A rash/hypersensitivity reaction is consistent with the known safety profile of beta-lactam antibiotics.
Causality	Possible to probable based on the available scenario; a formal validated scale should be completed using the full clinical record.

Safety note: This is a simulated educational case created for an ADR-identification project. Real patients require assessment by a qualified healthcare professional, and suspected allergic reactions should be documented in the patient's medical record.

5. Final Case Analysis & Reporting Plan

Patient-centered pharmacovigilance documentation

A. Case Summary

A 45-year-old female developed a new rash and itching together with nausea and dizziness approximately two days after starting amoxicillin. The close temporal relationship, symptom pattern and assumed improvement following withdrawal make an adverse drug reaction clinically plausible. The strongest signal in this case is the new cutaneous reaction; nausea and dizziness are less specific and should be evaluated for alternative causes.

B. Suspected Mechanism

The rash and pruritus may represent a hypersensitivity reaction to amoxicillin. The exact immunologic mechanism cannot be established from this limited case alone. Clinical severity must be assessed for features such as facial/tongue swelling, breathing difficulty, hypotension, mucosal involvement, blistering or systemic organ involvement.

C. Recommended Pharmacovigilance Data to Collect

Data field	What should be recorded
Patient details	Age, sex, relevant medical history, allergies, pregnancy status where applicable.
Drug exposure	Generic name, strength, dosage form, route, dose, frequency, indication, start/stop dates and batch details if available.
Concomitant therapy	All prescription, OTC, herbal and supplement products used around the event.
Reaction details	Exact onset date/time, signs and symptoms, severity, investigations, treatment and clinical outcome.
Dechallenge/rechallenge	Whether the drug was stopped and what happened afterward; record any previous exposure and reaction history.
Reporter information	Reporter type and contact/process details required by the applicable pharmacovigilance reporting system.

D. Conclusion

Based on the information supplied in the scenario, amoxicillin is the suspected medicine and the new rash with pruritus represents the principal suspected ADR. The case demonstrates the core pharmacovigilance workflow: identify the event, establish the medication timeline, evaluate alternative causes, assess dechallenge/rechallenge information, characterize seriousness and expectedness, and document the case for appropriate clinical and safety follow-up.

E. Student Learning Outcome

After completing this task, the learner should be able to recognize a potential ADR from a patient narrative, identify the suspected medicine, construct a clear event timeline, distinguish association from proof of causality, and prepare the essential information needed for pharmacovigilance documentation.

References for academic use

1. World Health Organization (WHO). Pharmacovigilance: ensuring the safe use of medicines.
2. Uppsala Monitoring Centre (UMC). WHO-UMC system for standardized case causality assessment.

3. International Council for Harmonisation (ICH). E2A: Clinical Safety Data Management — Definitions and Standards for Expedited Reporting.

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